Documentation for Perception Model Version 0

DARPA MATRICS PERCEPTION MODEL

Contract №

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1 INTRODUCTION

This documentation provides the details of version 0 of the perception model which is an Assume-Guarantee contracts based model captured in AADL with the AGREE annex []. This document details the contexts e.g. mission scenario, operator task, and the intended scope of applicastion in section 2. We also provide background information on the class of formal method tools that we use to capture and analyze.

Section 3 describes the formal model in details including the devised HSR protocol need to validate the operator portion of the formal model.

Section 4 describes the result of the security analysis and shows the CEX which represents a possible attack that the HMD failed to prevent.

Section 5 describes the future refinements or extensions of the model including more complexities on the device and operator.

2 BACKGROUND

2.1 Mission Scenario

The MATRICS team has defined a common mission scenario (with slight variations for each category). The mission centers around an HMD application used at border crossings and other security checkpoints for identifying people carrying suspicious packages. The system highlights the suspicious person by drawing a red bounding box around them, and the HMD operator (e.g., a checkpoint guard in an observation tower) must then visually confirm whether the highlighted person is indeed a threat or an attacker. If so, the guard will confirm the alert (which might then trigger ground personnel to intercept the person identified by the scan). If not, the guard will dismiss the alert, which will remove the bounding box. The guard can also alert to a suspicious person that was not identified by the scanning system. The guard will be positioned with a birds eye view of the surveillance area enhanced by a HMD display. People passing through the surveillance area can be moving in any direction and at any humanly possible speed. People may be carrying a bag or backpack. Suspicious packages will all look identical.

2.2 Mission Task

2.3 Perception Attack

For this scenario, we assume the adversary has not compromised the headset system i.e. they cannot directly manipulate what the headset displays to the user. The adversary party has enough control of the external environment to introduce different sources of intense light into that environment at any location. The intense light will be targeted at the general direction of the guard, causing temporary pupil constriction and compromised visual capabilities.

2.4 Scope of Application

2.5 Overarching Assumptions for Version 0

We make a few global assumptions which applies to the entire model. These assumptions are mostly made to simplify this first release of perception model. Some assumptions are made to either simplify the formal analysis or to justify the formal language used for the modeling and analysis.

1. There is only one attacker in the field-of-vision (FOV) of the HMD at any given time. This greatly simplifies this first version of the model as we can now abstract the presence of attacker in the FOV using a single Boolean variable. With multiple attackers at different locations within the FOV, complexities are introduced such as the location on the screen of the attacker and the different characteristics of the different attackers.

2. The HMD system is perfect (no false positives nor false negativese) in detecting an attacker i.e. marking the attacker with a red bounding box. This simplifies the behavior of the HMD system. False positives and false negatives will introduced in a future version of this model.

3. There is only one source of light attack and it is localized around the attacker. Multiple sources of attack from different locations in the environment could be introduced in future versions of the model.

4. The perception process in a human operator behaves like a reactive synchronous[4] process with a discrete-time clock. This assumption is made in the selection of lustre [3] (can think of it as a discrete subset of Simulink) as the underlying formal language for specifying the perception behavior of the human operator and formal reasoning/analysis using lustre-based model checkers [1, 2]. We hypothsize that this is a close enough abstraction of the human perception process i.e. at each sample period, it takes in a visual input, processes the visual input to extract some information (possible presence and location of attacker), store the extracted information in memory, and then starts over again on the next visual input.

5. The operator component operates on the same clock as HMD display i.e. synchronized and has the same sample period. This assumption is made to simplify the analysis in this first version of the model.

3 MODEL DETAILS

3.1 Assume-Guarantee join operator + device Model

1. device components
2. operator component, an abstraction of human perception
3. assume and guarantees
4. sources from the cognitive literature for estimating the parameter values int he assumption, guarantee formulas..
5. usage in designing filter, and perform security analysis..

3.2 HSR Protocol for Validting the Operator Component

We plan conduct human subject experimentations to validate the operator component. To validate the operator model, we have devised the following Subject Research Protocol

Description: In an environment where the ambient lighting is constant, there is 1 target in the field of view. Target walks along path starting from step 1 to 5. A bounding box around the target appears between step 2- 4. The timing of when a bounding box is randomized to prevent participant learning effect. The location of the bounding box relative to the target is constant, but its appearance is also randomized to minimize learning effect. The design choices of the bounding box incorporates current Helmet-Mounted symbology concepts for degraded environments (Cheung et al., 2015). Participants Instructions: Bounding boxes may or may not appear around a pedestrian. If you see a bounding box, press a button as soon as you see it. Measure: Participant performance in

1. true and false positives for identifying existence of bounding box
2. delays in correct identification.

Baseline Condition: No attack, target walks along path. Bounding box appears. Attack Condition: Glare attack: Target turns on a headlamp somewhere along the walking path. Luminance level of headlamp matches or is greater than bounding box. Headlamp is pointing at HMD. Bounding box appears Mitigation Condition: HMD modifies the stimuli by introducing a regional diming filter around the high luminance area produced in the headlamp. See example of auto-diming glass (https://www.gentextech.com/dimmable-glass.html). Timing: Six combinations with an estimated 15 seconds per run. Each subject performs multiple sets of the full combination for a total of 10 minute of subject task time. We may choose to add variations in ambient lighting conditions should power analysis permits. Hypotheses for Baseline, Attack, & Mitigation conditions:

1. Baseline: Participants will correctly identify symbology (DV1) at 90% with delays (DV2) <=600ms (+/- 60ms).

2. Baseline DV1 estimates consider the presence and absence of factors specific to our stimuli that affect human error rates (Remington & Willaims, 1985).

3. Baseline DV2 estimates delay leverages findings in the analogous visual-motor task of a cricket batter perceiving the ball release to judging the futural arrival point of the ball (Sarpeshkar & Mann, 2011).

4. Attack: Participants will correctly identify symbology (DV1) at 80% with delays (DV2) >= 1sec.

The expected performance decrement in delay is based on a hypothesized function of the following features:

1. Attack light source of 100 lumens such as from a low cost flashlight.

2. A maximum pupil size change of 1.15mm for a maximum luminance change of 10.90cd/m2 (standard deviation 20.04) (see Table 1 and 2, Sulutvedt et al., 2021)

3. 1mm increase in pupil diameter corresponds to an average increase in the perceptual experience of 2.09 cd/m2 in brightness (Sulutvedt et al., 2021)

Mitigation: Return to baseline for both DV1 and DV2 with a 10% performance decrement.

Independent Variables

4 ANALYSIS

4.1 AGREE analysis

1. verification monolithically..

2. CEX, what are they, and how are they useful for our intended scope of application which is a security assessment.

3. translation into lustre [] which is checked by infinite-state model checker

4. kind2, jkind

5. results

5 FUTURE REFINEMENTS

Potential Extensions

1. Lighting strobe lights/different frequencies
2. Occlusions
3. Shadows
4. Color
5. Different modalities: auditory, auditory+visual

Relaxation of the overaching assumption: multiple targets, dynamic targets moving across the FOV,

Model of human decision making in perception.. you plan to see, and see to plan [].

References

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[3] Nicolas Halbwachs. A synchronous language at work: the story of lustre. *Formal Methods for Industrial Critical Systems: A Survey of Applications*, pages 15–31, 2012.

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